

ConvoShop

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Contents

1. Introduction	6
1.1 Problem Statement	6
1.2 Scope	6
1.3 Modules	7
1.3.1 LLM Interaction Module	7
1.3.2 Web Scraping Agent Module	7
1.3.3 Product Comparison Module	8
1.3.4 Sentiment Analysis Module	8
1.3.5 Review Aggregation & Analysis Module	8
1.3.6 Price Monitoring & Alert Module	9
1.3.7 Data Storage & Management Module	9
1.3.8 Security & Web Scraping Control Module	9
1.3.9 User Management & Preferences Module	9
1.3.10 Error Handling & Notifications Module	10
2. Project Requirements	11
2.1 Use-case/Event Response Table/ Storyboarding	11
2.2 Functional Requirements	12
2.2.1 LLM Interaction Module	12
2.2.2 Web Scraping Agent Module	12
2.2.3 Product Comparison Module	12
2.2.4 Sentiment Analysis Module	13
2.2.5 Review Aggregation & Analysis Module	13
2.2.6 Price Monitoring & Alert Module	13
2.2.7 Data Storage & Management Module	14
2.2.8 Security & Web Scraping Control Module	14
2.2.9 User Management & Preferences Module	14
2.2.10 Error Handling & Notifications Module	14
2.3 Non-Functional Requirements	15
2.3.1 Performance	15

2.3.2 Scalability.....	15
2.3.3 Security.....	15
2.3.4 Usability	15
2.3.5 Reliability	15
2.3.6 Maintainability	16
2.3.7 Compliance.....	16
2.4 Domain Model.....	16
3. System Overview.....	17
3.1 Architectural Design	17
3.2 Design Models	18
3.2.1 Data Flow Diagrams	18
3.2.2 Activity Diagram.....	21
3.2.3 System Level Sequence Diagrams.....	22
3.2.4 JSON Schema.....	27

List of Figures

FIGURE 1 DOMAIN MODEL	16
FIGURE 2 ARCHITECTURE DESIGN.....	17
FIGURE 3 DATA FLOW DIAGRAM LEVEL 0.....	18
FIGURE 4 DATA FLOW DIAGRAM LEVEL 1.....	19
FIGURE 5 DATA FLOW DIAGRAM LEVEL 2.....	20
FIGURE 6 ACTIVITY DIAGRAM.....	21
FIGURE 7 SSD USE CASE 1	22
FIGURE 8 SSD USE CASE 2	23
FIGURE 9 SSD USE CASE 3	24
FIGURE 10 SSD USE CASE 4	25
FIGURE 11 SSD USE CASE 5	26
FIGURE 12 SSD USE CASE 6	27

List of Tables

TABLE 1 EVENT RESPONSE TABLE 11

Chapter 1

1. Introduction

Our goal is to develop ConvoShop, a project aimed at helping users find products across various e-commerce platforms through an easy-to-use conversational interface and a browser extension. The system is designed to address the challenges of time-consuming and complicated product comparisons on different websites. By utilizing web scraping and AI within a conversational framework, the system will simplify the process for users to discover and compare products in real-time, offering personalized recommendations based on their preferences. The project will enable users to search for products by engaging with the conversational interface, which will collect data from multiple platforms and recommend the best deals

1.1 Problem Statement

Shopping online can be challenging due to the vast number of products and websites available. Users frequently find themselves spending considerable time sifting through various sites to locate the best prices, reviews, and options. This can be a tedious process, often resulting in missed chances for better deals or products. Our system is designed to address this issue by providing a conversational interface that searches multiple websites, compares products, and offers suggestions all in one convenient location. It will help users save time and effort by automating the product search and comparison process while delivering personalized recommendations

1.2 Scope

The scope of this project covers the development of a conversational AI shopping assistant. The main functionalities include:

1. **Web Scraping:** The system will collect data from multiple e-commerce websites. This will give users access to real-time product information by displaying web elements from different platforms.

2. **Conversational Interface:** Users will interact with the system through a text-based chat interface. This interface will use natural language processing to understand user queries and preferences.
3. **Product Comparison:** The system will compare products across different platforms. It will look at prices, features, and availability.
4. **Sentiment Analysis:** The system will analyze user reviews to predict product sentiment. This will help users understand product quality and customer satisfaction.
5. **Personalized Recommendations:** Based on user interactions and preferences, the system will suggest relevant products.
6. **Real-time Data Integration:** The system will provide up-to-date information on prices, availability, and reviews.
7. **Cross-Platform Analysis:** Users will be able to compare products from different e-commerce sites in one place.
8. **Responsive User Interface:** The system will have a user-friendly interface that works well on both desktop and mobile devices.

1.3 Modules

1.3.1 LLM Interaction Module

Function: Manages conversational interactions with the user, taking input queries and generating appropriate responses.

Tasks:

- Process user queries.
- Handle clarification requests in case of ambiguous or incomplete input.
- Communicate with other modules to retrieve necessary data (product details, comparisons, etc.)

1.3.2 Web Scraping Agent Module

Function: Scrapes product data from selected online stores based on user queries.

Tasks:

- Real-time retrieval of product information like prices, reviews, sales, and availability.

- Handle multiple platforms, ensuring that data from different stores is aggregated.
- Retry scraping operations in case of failure or delay.

1.3.3 Product Comparison Module

Function: Compares product data across multiple platforms.

Tasks:

- Aggregate product data from different sources (platforms).
- Identify key product metrics (price, ratings, sales) and present them to the user.
- Handle cases where some platforms fail to respond or have outdated data.

1.3.4 Sentiment Analysis Module

Function: Analyzes product reviews to determine the overall sentiment and helps users decide whether the product is worth buying.

Tasks:

- Perform sentiment analysis on user-generated reviews from various platforms.
- Aggregate sentiment ratings into an overall score (positive, neutral, negative).
- Notify the user if sentiment analysis fails or if review data is incomplete.

1.3.5 Review Aggregation & Analysis Module

Function: Aggregates both user and professional reviews to generate comprehensive product ratings.

Tasks:

- Collect and aggregate reviews from different sources (both user and expert reviews).
- Assign overall ratings to products based on review aggregation.
- Provide detailed review insights and handle professional-review analysis when available.

1.3.6 Price Monitoring & Alert Module

Function: Tracks product prices over time and alerts users when price changes occur.

Tasks:

- Monitor prices of selected products in real-time or periodically.
- Notify users if there are significant changes in price or ratings of the tracked products.
- Store price history for future reference and comparison.

1.3.7 Data Storage & Management Module

Function: Manages storage of product information, user preferences, and data retrieved via scraping.

Tasks:

- Store product data, including prices, reviews, sales, etc.
- Keep track of user preferences and selections (e.g., monitored products).
- Manage data caching to avoid repetitive scraping of the same information.

1.3.8 Security & Web Scraping Control Module

Function: Ensures safe operation of web scraping agents and prevents access to harmful or malicious websites.

Tasks:

- Monitor web scraping activities for security threats.
- Ensure compliance with site terms of service and prevent scraping of restricted data.
- Implement rate-limiting and scraping control mechanisms to prevent system overload or blocking by websites.

1.3.9 User Management & Preferences Module

Function: Handles user profiles, preferences, and personalized settings.

Tasks:

- Allow users to set product preferences, track specific products, or receive alerts.
- Manage user authentication and settings, like notifications for price changes.
- Personalize product suggestions based on past user queries and interactions.

1.3.10 Error Handling & Notifications Module

Function: Manages all system errors and informs users about potential issues in real-time.

Tasks:

- Catch and log errors from all modules (scraping failures, analysis engine failures, etc.).
- Provide user notifications in case of system delays, missing data, or partial results.
- Handles retry logic for failed web scraping operations.

Chapter 2

2. Project Requirements

2.1 Use-case/Event Response Table/ Storyboarding

Event	Response	Priority	Responsible Party	Expected Outcome	Escalation	Timeframe
User Input Ambiguity	Ask the user for clarification or suggest relevant products based on partial input	Medium	LLM Interaction Module	User clarifies or accepts suggestion	Notify developer if frequent	Immediate
Real-time Search Timeout	Notify the user about the delay and retry the web scraping request	High	Web Scraping Agent	Product data retrieved or retry after delay	Escalate to Web Agent Manager	30 seconds
Failed Multi-platform Comparison	Display available results and notify user about platforms that failed to respond	Medium	Web Scraping Agent	User receives partial comparison with status of other sites	Retry after delay	1 minute
Sentiment Analysis Engine Failure	Display review content without sentiment analysis and inform user of the issue	Low	Sentiment Analysis Engine	User still receives reviews, but sentiment rating is absent	Restart sentiment engine	5 minutes
No Professional Reviews Available	Notify the user and provide aggregated user reviews instead	Low	Review Analysis Module	User receives alternative review data	Suggest professional sources	Immediate
Price Monitoring Data Missing	Display available price data and inform user about missing or outdated information	Medium	Price Monitoring System	User is aware of missing product metrics	Escalate if persistent	Immediate
Product Rating Aggregation Failure	Display user reviews and ratings without aggregation, notify user	Low	Sentiment Analysis Module	User still receives individual reviews and ratings	Escalate to developer	Immediate
Price/Ratings Updated on Monitored Products	Automatically notify the user about price or rating changes on selected products	High	Price Monitoring System	User receives updates about tracked products	N/A	Immediate

Table 1 Event Response Table

2.2 Functional Requirements

2.2.1 LLM Interaction Module

- **Description:** The system shall allow users to input product-related queries and return relevant product information.
 - **Priority:** High
 - **Details:**
 - The module shall handle ambiguous or incomplete input by asking for clarification or providing suggestions.
 - It must integrate with LangChain to enhance conversational abilities.
 - The module shall send product data requests to other modules such as web scraping, sentiment analysis, etc.
-

2.2.2 Web Scraping Agent Module

- **Description:** The system shall retrieve product data (prices, reviews, sales, availability) from multiple e-commerce platforms.
 - **Priority:** High
 - **Details:**
 - It must scrape from multiple sources such as Daraz, Amazon, and eBay.
 - The module shall retry scraping in case of timeouts or failures.
 - Scraped data must be passed to comparison, sentiment analysis, and UI modules.
-

2.2.3 Product Comparison Module

- **Description:** The system shall aggregate product data from various platforms and present it to users.
- **Priority:** Medium
- **Details:**
 - It must handle missing or outdated data and notify users of issues while displaying available results.
 - The module shall compare products based on price, reviews, and sales metrics.

2.2.4 Sentiment Analysis Module

- **Description:** The system shall perform sentiment analysis on product reviews and assign an overall rating (positive, neutral, negative).
- **Priority:** Medium
- **Details:**
 - The module shall aggregate sentiment ratings from multiple platforms and provide a comprehensive score.
 - It should continue to provide reviews in case of analysis failure while notifying the user.

2.2.5 Review Aggregation & Analysis Module

- **Description:** The system shall collect both user and professional reviews from different platforms.
- **Priority:** Medium
- **Details:**
 - It must generate an overall product rating based on aggregated reviews.
 - The module shall handle scenarios where no professional reviews are available.

2.2.6 Price Monitoring & Alert Module

- **Description:** The system shall track price changes for user-selected products.
 - **Priority:** High
 - **Details:**
 - It shall notify users when there is a significant price drop or a change in product ratings.
 - The module shall store price history for future reference and comparison.
-

2.2.7 Data Storage & Management Module

- **Description:** The system shall store product data, user preferences, search histories, and monitoring lists.
 - **Priority:** Medium
 - **Details:**
 - It must handle large datasets using MongoDB and Vector DB for efficient retrieval.
 - The system shall cache data to prevent repetitive scraping of the same product information.
-

2.2.8 Security & Web Scraping Control Module

- **Description:** The system shall ensure web scraping agents do not access malicious or restricted websites.
 - **Priority:** High
 - **Details:**
 - It shall monitor security threats and handle rate-limiting or IP blocking by e-commerce sites.
-

2.2.9 User Management & Preferences Module

- **Description:** The system shall implement login/logout functionality and allow users to set preferences for product monitoring.
 - **Priority:** Medium
 - **Details:**
 - User accounts shall be secured, and preferences stored for personalized suggestions.
-

2.2.10 Error Handling & Notifications Module

- **Description:** The system shall catch and log errors across all modules (LLM failures, scraping errors, sentiment analysis failures, etc.).
- **Priority:** High
- **Details:**
 - The system shall notify users in case of delays, missing data, or partial results.

- It must implement retry mechanisms for failed web scraping or sentiment analysis tasks.

2.3 Non-Functional Requirements

2.3.1 Performance

- The system should return search results and product comparisons within 2-3 seconds on average.
- Web scraping and sentiment analysis operations must complete within 5-10 seconds depending on data volume.

2.3.2 Scalability

- The system should scale to handle multiple concurrent user requests and web scraping operations without significant performance degradation.
- MongoDB and Vector DB should support large datasets with efficient indexing and retrieval.

2.3.3 Security

- User data must be encrypted both at rest and in transit.
- Web scraping agents should avoid violating the terms of service of target websites and respect rate limits.
- Implement secure authentication (OAuth or JWT) for user login and personalized features.

2.3.4 Usability

- The conversational UI must be intuitive, guiding users through product searches and comparisons seamlessly.
- The mobile app must have feature parity with the web app and maintain a consistent user experience across devices.

2.3.5 Reliability

- The system should have 99.9% uptime, ensuring consistent availability of product search, comparison, and monitoring features.
- Web scraping agents must have a fallback system to handle platform downtime or data unavailability.

2.3.6 Maintainability

- The system must have modular and well-documented code for ease of updates and integration of new platforms or features.
- Support automated tests and monitoring for easy identification of issues or bottlenecks.

2.3.7 Compliance

- Ensure compliance with data privacy regulations such as GDPR if storing user data.
- Web scraping agents should comply with website terms and avoid scraping restricted or protected data.

2.4 Domain Model

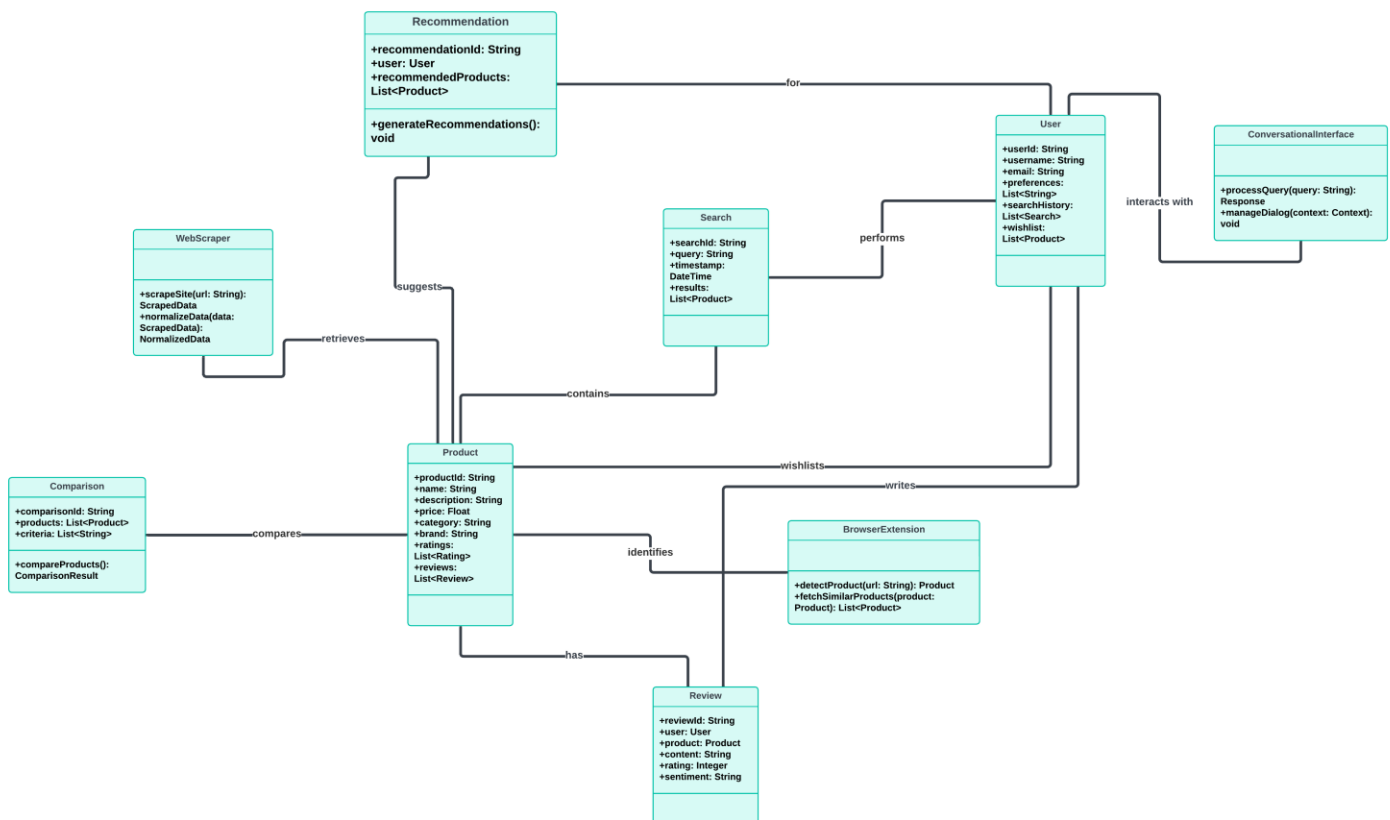


Figure 1 Domain Model

Chapter 3

3. System Overview

3.1 Architectural Design

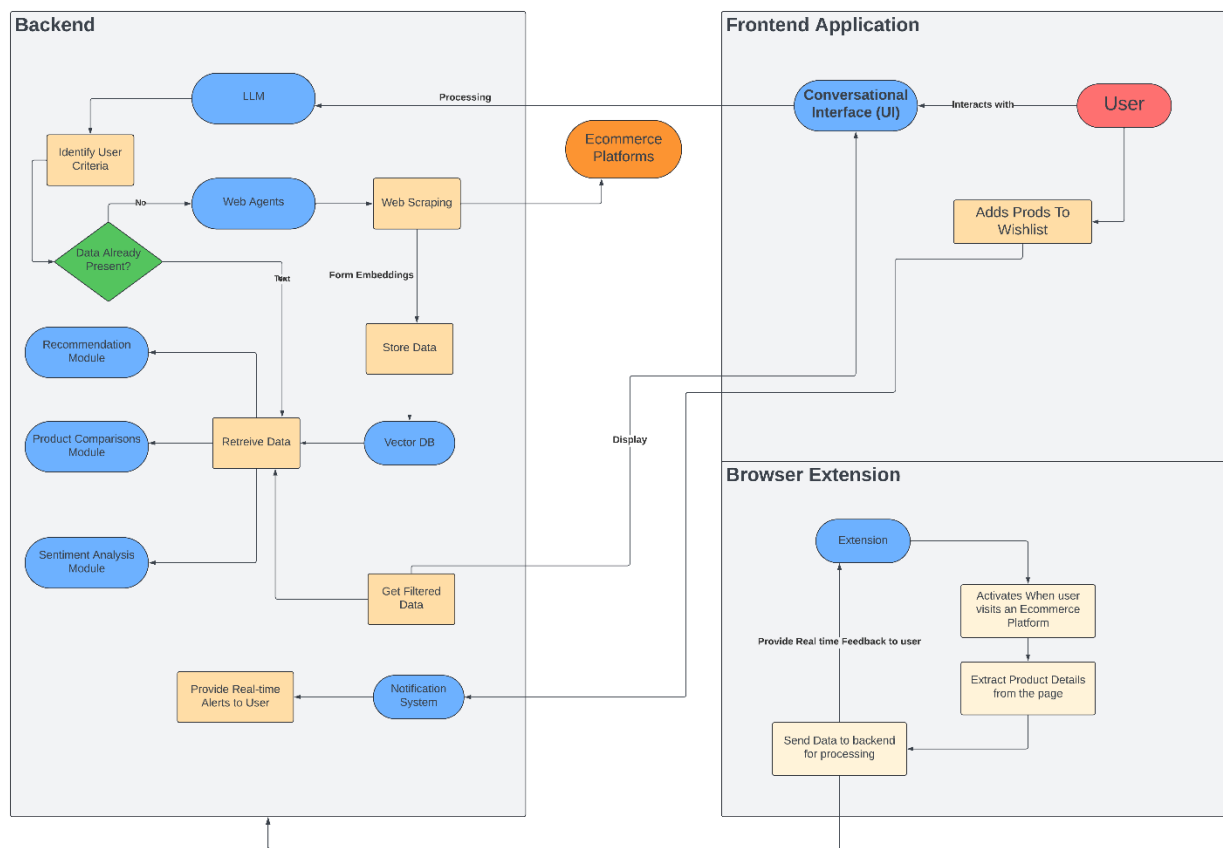


Figure 2 Architecture Design

3.2 Design Models

3.2.1 Data Flow Diagrams

DFD Level 0

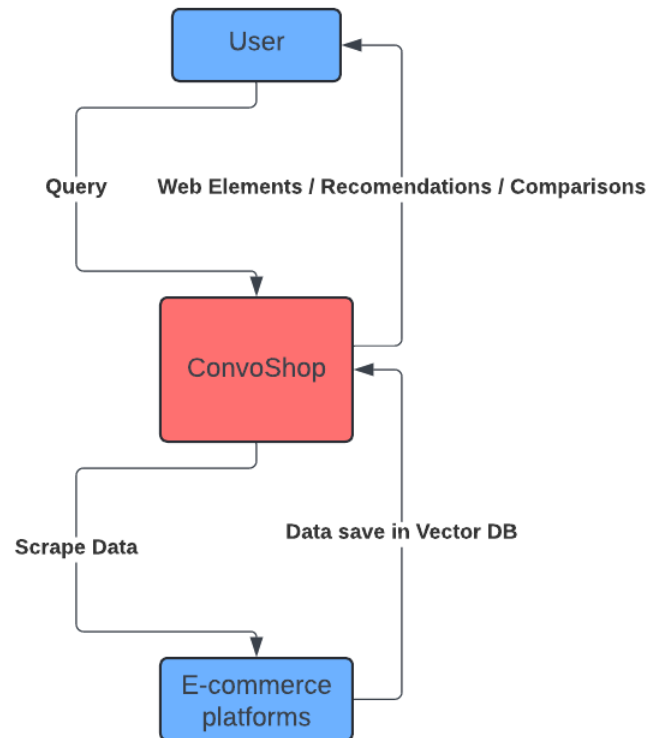


Figure 3 Data Flow Diagram Level 0

DFD Level 1

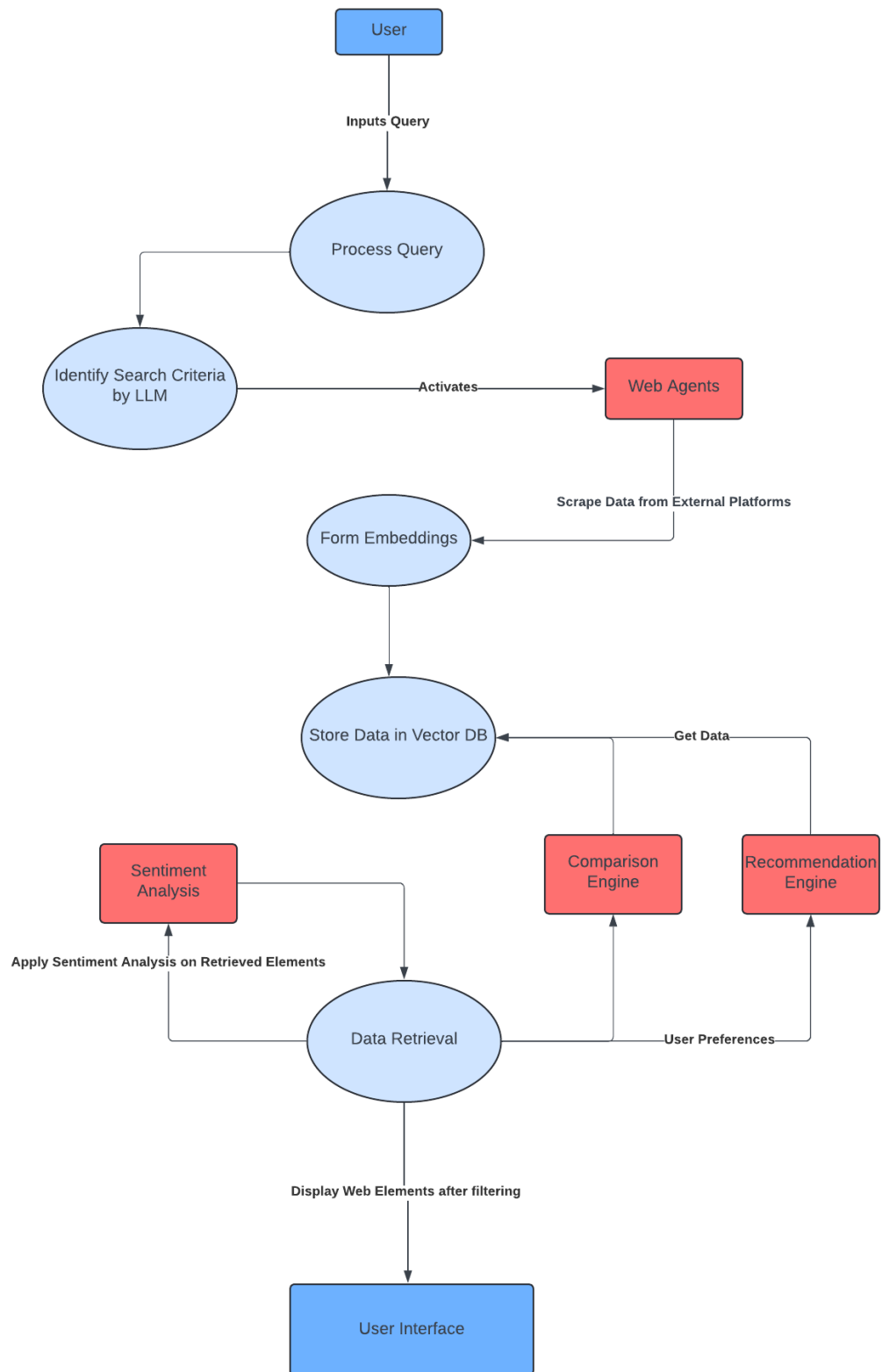


Figure 4 Data Flow Diagram Level 1

DFD Level 2

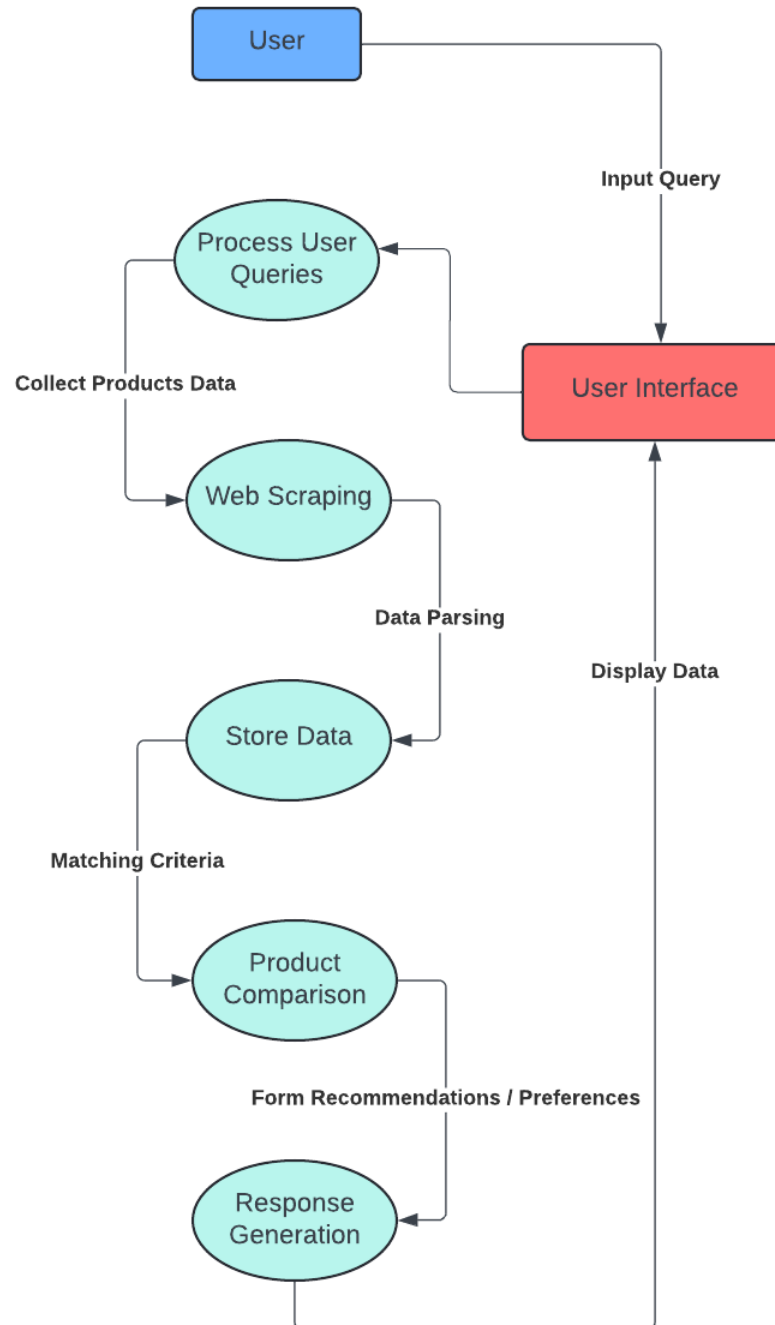


Figure 5 Data Flow Diagram Level 2

3.2.2 Activity Diagram

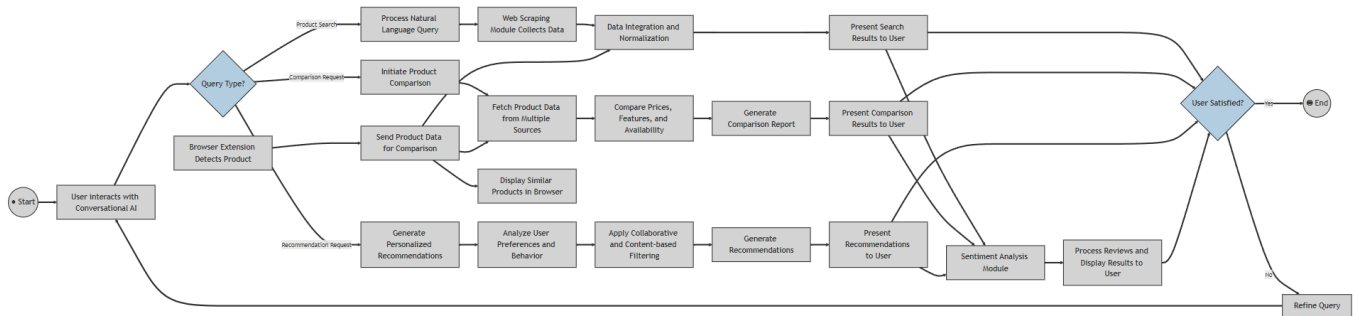


Figure 6 Activity Diagram

3.2.3 System Level Sequence Diagrams

1. Product Search and Comparison

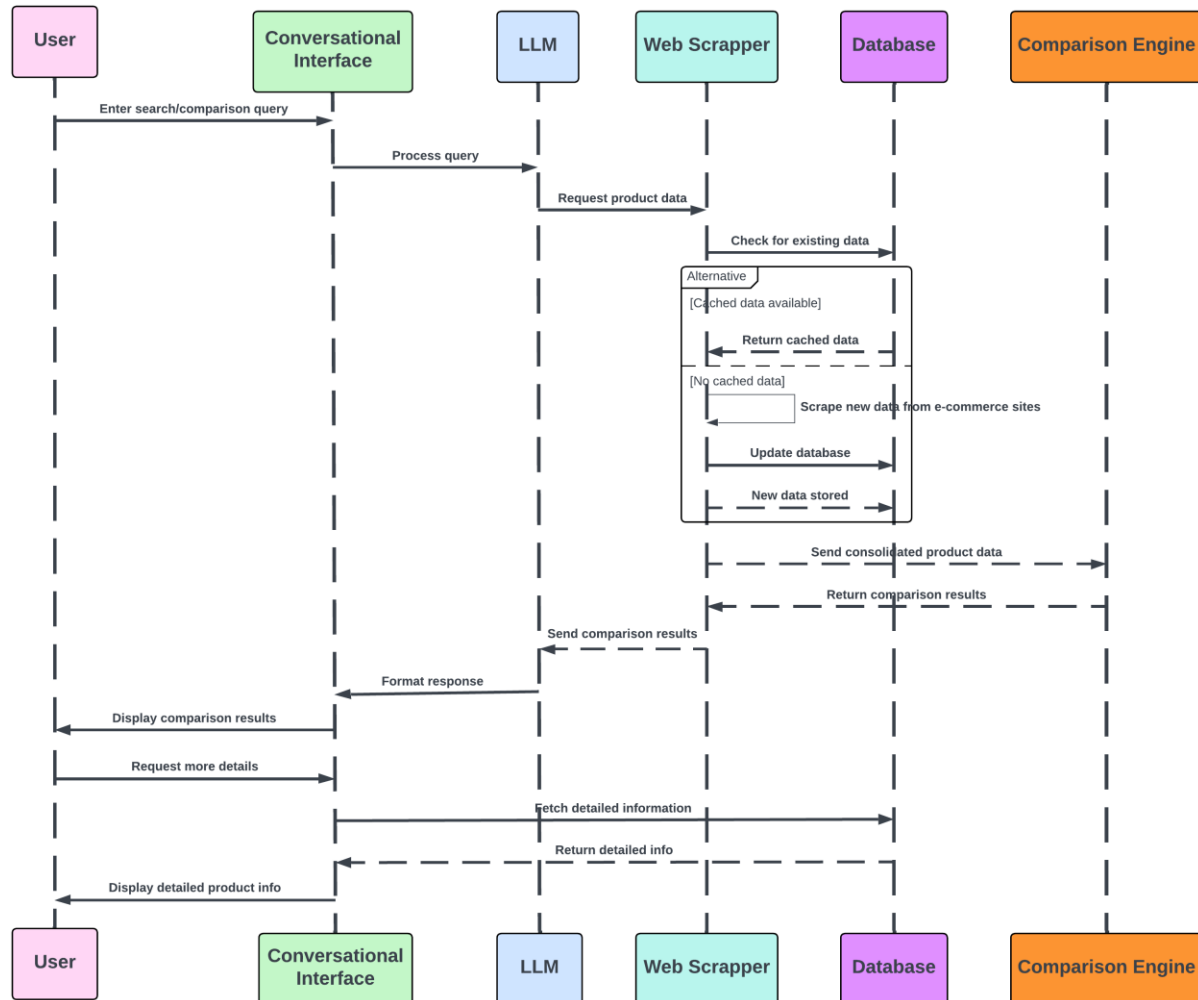


Figure 7 SSD use case 1

2. Recommendation and Sentiment Analysis

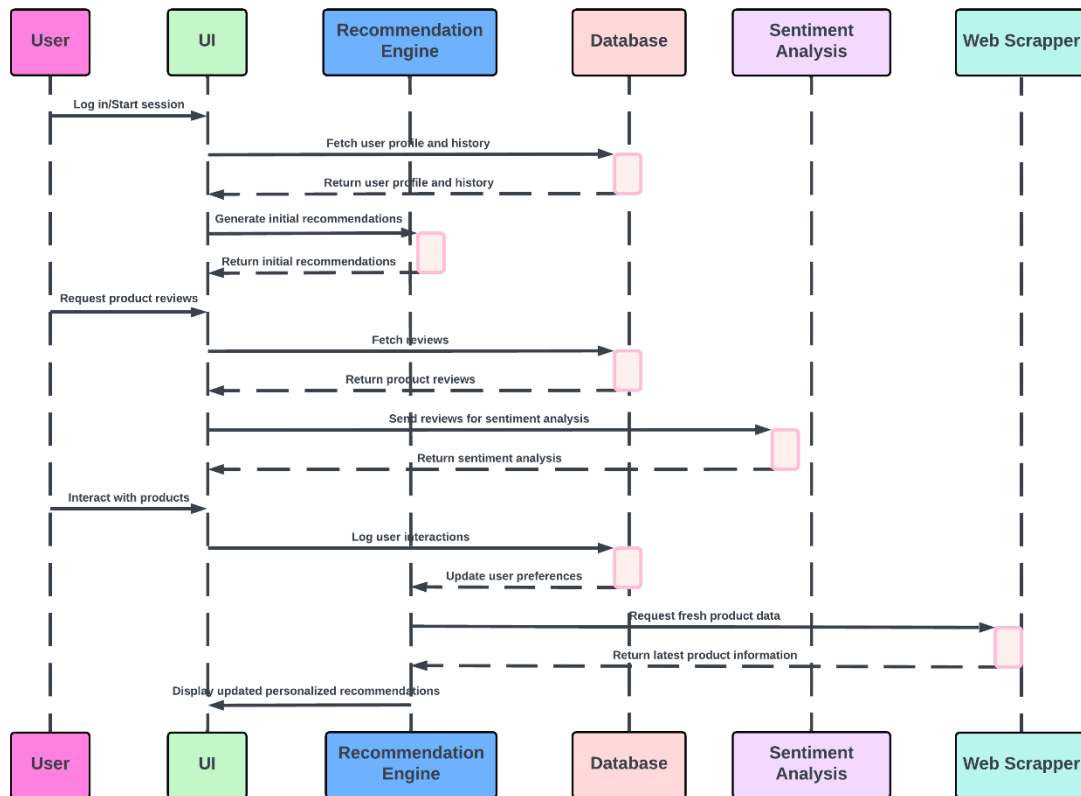


Figure 8 SSD use case 2

3. Cross Platform Shopping with Filtering

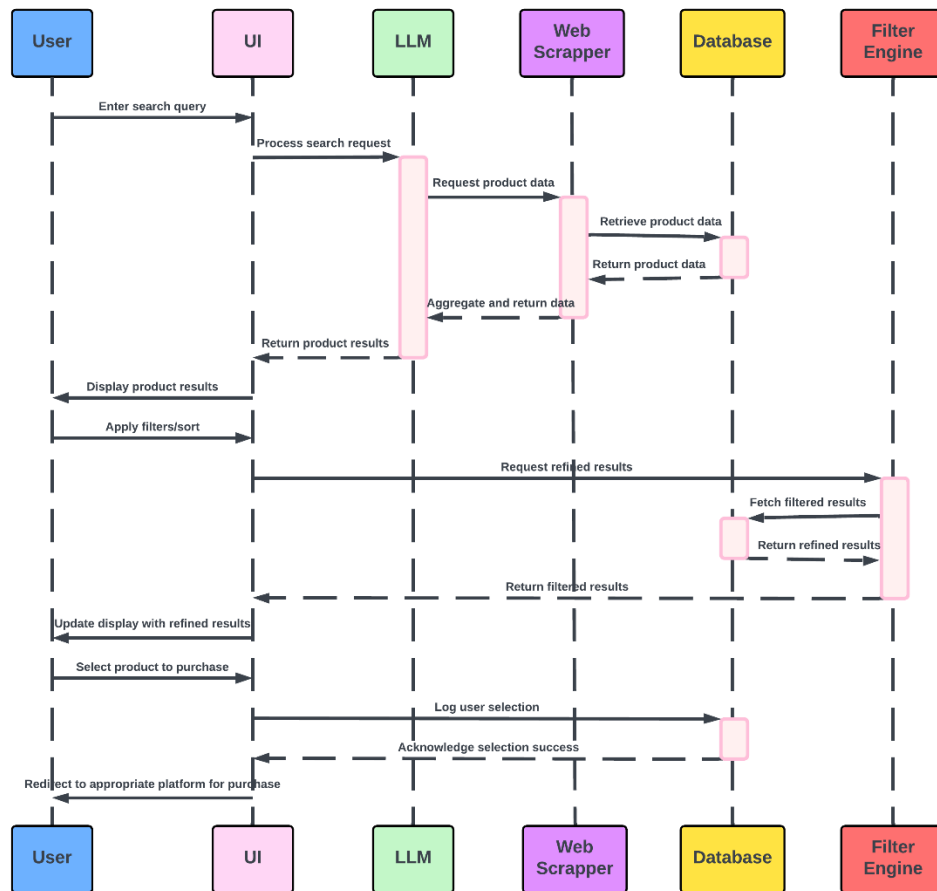


Figure 9 SSD use case 3

4. Real Time Price Monitoring and Wishlist

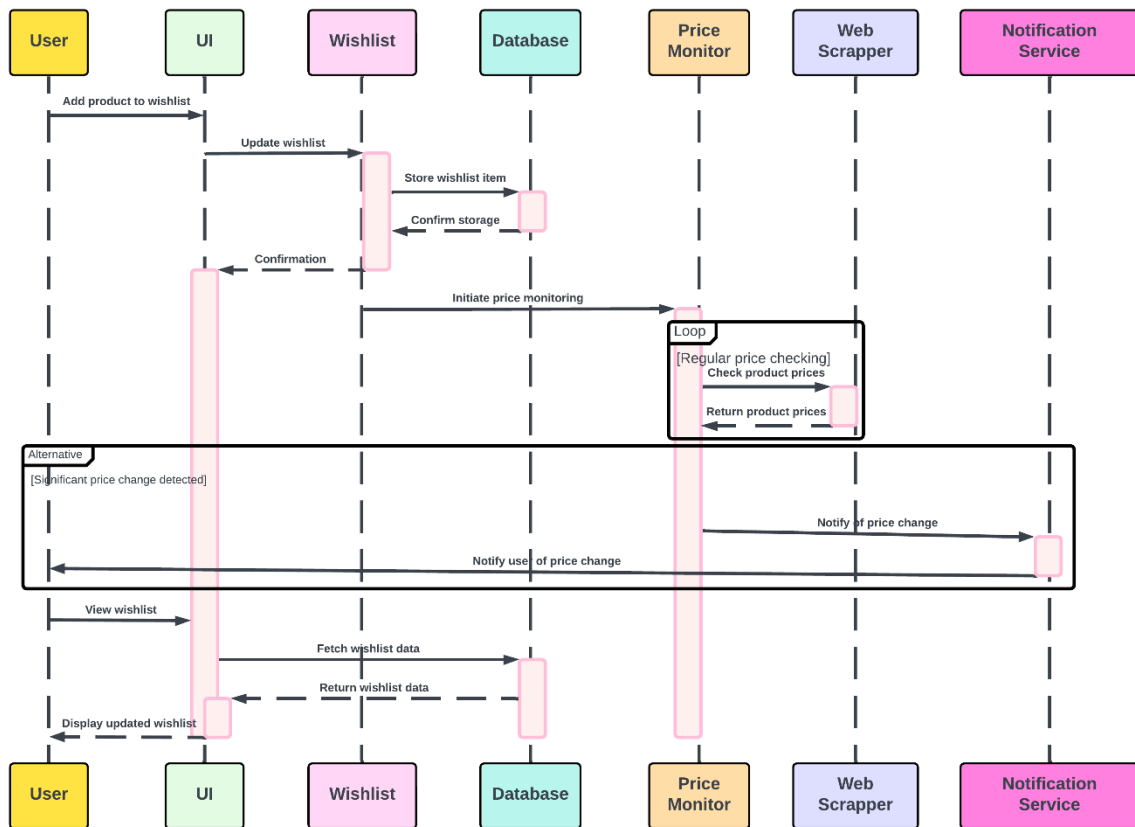


Figure 10 SSD use case 4

5. Browser Extension for Product Comparison

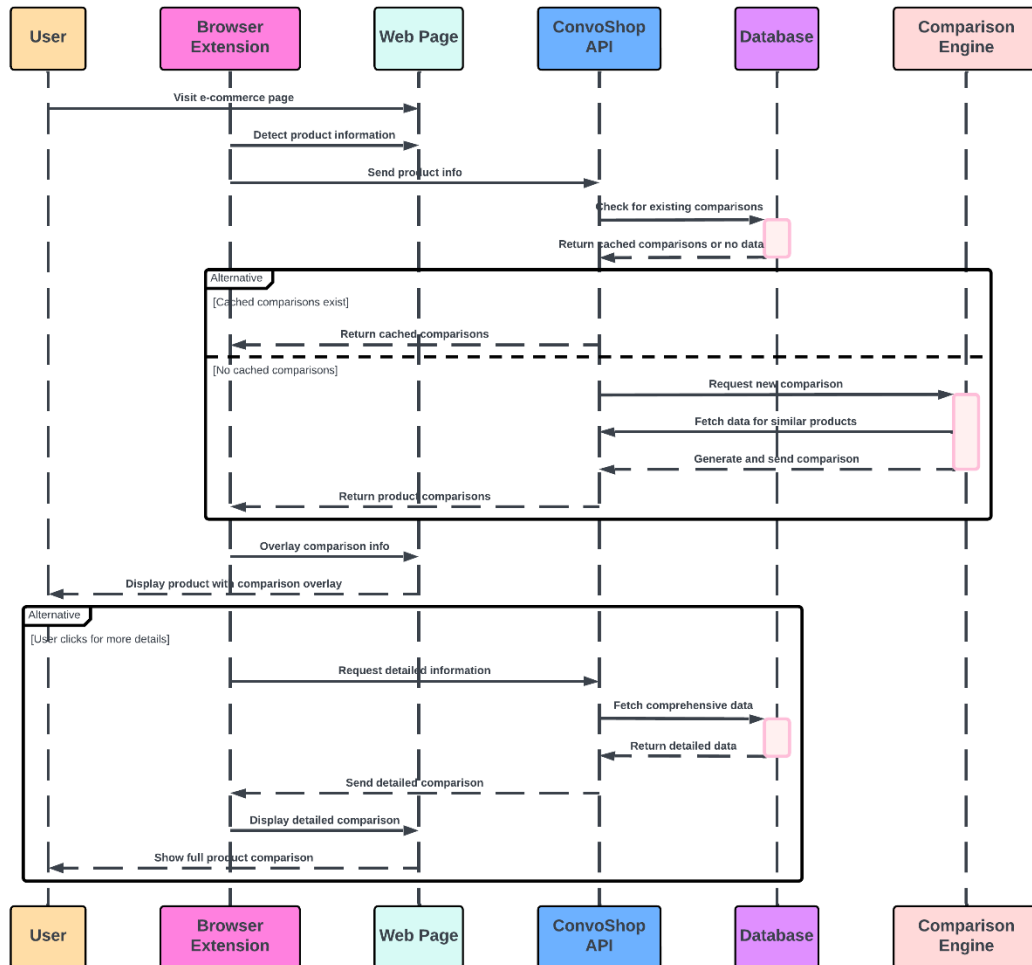


Figure 11 SSD use case 5

6. Math Agent for Price Calculation and Comparisons

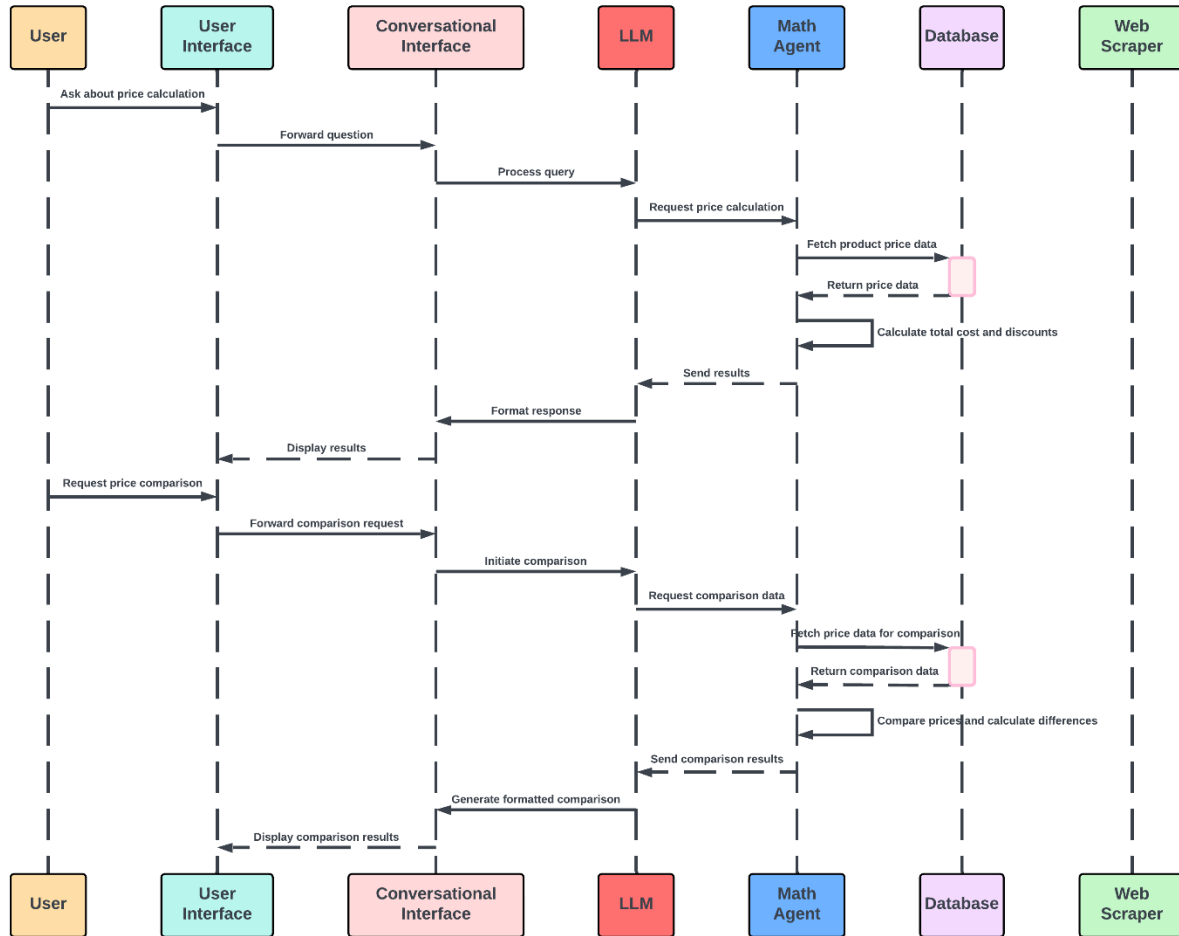


Figure 12 SSD use case 6

3.2.4 JSON Schema

```

{
  "$schema": "http://json-schema.org/draft-07/schema#",
  "title": "ConvoShop Data Schema",
  "type": "object",
  "properties": {

```

```
"users": {
  "type": "array",
  "items": {
    "type": "object",
    "properties": {
      "userId": { "type": "string" },
      "username": { "type": "string" },
      "email": { "type": "string", "format": "email" },
      "preferences": {
        "type": "array",
        "items": { "type": "string" }
      },
      "searchHistory": {
        "type": "array",
        "items": { "$ref": "#/definitions/search" }
      },
      "wishlist": {
        "type": "array",
        "items": { "$ref": "#/definitions/product" }
      }
    },
    "required": ["userId", "username", "email"]
  },
  "products": {
    "type": "array",
    "items": { "$ref": "#/definitions/product" }
  },
  "reviews": {
    "type": "array",
    "items": { "$ref": "#/definitions/review" }
  },
```

```
"comparisons": {
  "type": "array",
  "items": { "$ref": "#/definitions/comparison" }
},
"recommendations": {
  "type": "array",
  "items": { "$ref": "#/definitions/recommendation" }
}
},
"definitions": {
  "product": {
    "type": "object",
    "properties": {
      "productId": { "type": "string" },
      "name": { "type": "string" },
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        "items": {
          "type": "object",
          "properties": {
            "userId": { "type": "string" },
            "value": { "type": "number", "minimum": 1, "maximum": 5 }
          }
        }
      }
    }
  },
  "reviews": {
    "type": "array",
    "items": { "$ref": "#/definitions/review" }
  }
}
```

```
    },
    "sourceWebsite": { "type": "string" },
    "lastUpdated": { "type": "string", "format": "date-time" }
  },
  "required": ["productId", "name", "price", "sourceWebsite"]
},
"review": {
  "type": "object",
  "properties": {
    "reviewId": { "type": "string" },
    "userId": { "type": "string" },
    "productId": { "type": "string" },
    "content": { "type": "string" },
    "rating": { "type": "integer", "minimum": 1, "maximum": 5 },
    "sentiment": { "type": "string", "enum": ["positive", "neutral", "negative"] },
    "date": { "type": "string", "format": "date-time" }
  },
  "required": ["reviewId", "userId", "productId", "content", "rating"]
},
"search": {
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  "properties": {
    "searchId": { "type": "string" },
    "query": { "type": "string" },
    "timestamp": { "type": "string", "format": "date-time" },
    "results": {
      "type": "array",
      "items": { "$ref": "#/definitions/product" }
    }
  },
  "required": ["searchId", "query", "timestamp"]
},
```

```
"comparison": {
  "type": "object",
  "properties": {
    "comparisonId": { "type": "string" },
    "products": {
      "type": "array",
      "items": { "$ref": "#/definitions/product" },
      "minItems": 2
    },
    "criteria": {
      "type": "array",
      "items": { "type": "string" }
    },
    "result": { "type": "object" }
  },
  "required": ["comparisonId", "products", "criteria"]
},
"recommendation": {
  "type": "object",
  "properties": {
    "recommendationId": { "type": "string" },
    "userId": { "type": "string" },
    "recommendedProducts": {
      "type": "array",
      "items": { "$ref": "#/definitions/product" }
    },
    "reason": { "type": "string" },
    "timestamp": { "type": "string", "format": "date-time" }
  },
  "required": ["recommendationId", "userId", "recommendedProducts", "timestamp"]
}
}
```